46. Permutations

dfs([1,2,3], path = [])

([2,3], [1]) dfs([1,3], [2]) dfs([1,2], [3])

([3], [1,2]) (2, [1,3])

([], [1,2,3])

TC: $O(n \cdot n!)$

SC: $O(n \cdot n!)$ Output

$O(n^2)$ Auxillary

Recursion depth: n

At lvl k

Auxillary Stack

[1, 2, 3]	[]
[2,3]	[1]
[3]	[1, 2]
[]	[1, 2, 3]